rudimentary engineering, unstable control mechanisms, and unreliable communication systems. While they proved that pilotless flight was possible, their limited performance prevented them from achieving large-scale operational success.

- *Limited range and endurance*
  First-generation drones suffered from short flight durations and limited operational range, largely due to weak radio transmitters and small onboard fuel capacities. Early radio control systems could only transmit signals over a few kilometers, forcing drones to remain within the operator's line of sight. Additionally, inefficient engines and lightweight airframes restricted their endurance to just a few minutes of flight, making long-distance missions impossible.
- *Lack of stability and control precision*
  The absence of onboard stabilization systems, sensors, or autopilots made these drones difficult to maneuver. Control inputs were transmitted manually by the operator, often with noticeable delays and inaccuracies. Even slight wind disturbances could cause severe instability or crashes. Without gyroscopic or electronic feedback systems, maintaining smooth and controlled flight was a major challenge.
- *Primitive communication systems*
  Early drones relied on basic radio frequency (RF) control, which was prone to signal interference, low power output, and limited bandwidth. The lack of secure or multi-channel communication often led to loss of control mid-flight. Additionally, the analog transmission systems of the time could not support complex commands or telemetry data, restricting drones to simple directional control.
- *Minimal payload capacity*